

R2-D2 Desert Look Development and Lighting

Lighting Technical Director and Lighting Artist Case Study



Final cinematic frame • Katana and RenderMan physical render with a controlled Nuke finish

This personal portfolio project takes an existing R2-D2 asset from Maya and Alembic ingestion through Katana scene assembly, RenderMan look development and lighting, multilayer output, and a controlled Nuke finish. My objective was to make the character feel physically present in a dry canyon while preserving the familiar white, blue, and aluminium design.

PROJECT	DETAILS
Role	Lighting TD, Lighting Artist, Look Development, Compositing, Pipeline Automation
Pipeline	Maya and Alembic → Katana → RenderMan and OSL → Nuke
Format	1920 × 816 • 2.35:1 • 60 mm camera
Deliverables	Physical beauty, AOVs, Cryptomatte, denoised EXR, Nuke comp, comparisons

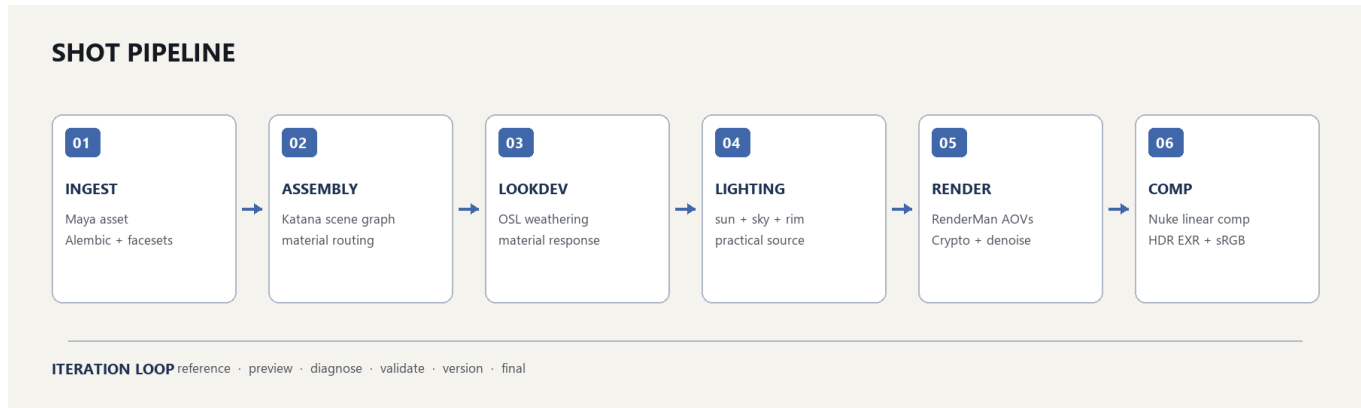
My contribution

- Scene ingestion, material routing, shader development and environment integration.
- Procedural weathering, shot lighting, RenderMan AOV design, denoising and Nuke compositing.
- Python automation, versioning, diagnostic rendering and technical validation. The R2-D2 model was an existing asset.

The challenge and production plan

The starting point was recognizable, but the asset read as too clean and separate from the environment. The dome felt grey rather than used aluminium, the white body lacked localized desert exposure, and an early procedural rock scatter made the floor visually busy. Character and canyon also occupied similar midtone values, which reduced depth.

I kept the camera and broad composition stable while solving these issues in the physical render first. Nuke was reserved for measured integration, depth, the practical-light bloom, and final film response.



A versioned pipeline with a diagnostic feedback loop between artistic decisions and technical validation.

Success criteria

- Keep the classic white, blue and silver read while adding believable desert exposure.
- Use motivated lighting and material response to integrate the asset before relying on comp.
- Preserve camera, geometry, UVs and approved assignments across every lookdev iteration.
- Make every selective adjustment available as an AOV, mask, light group or Cryptomatte selection.
- Deliver editable scenes, HDR data, review images and evidence that the source state remained intact.

Step 1 Scene ingestion and asset audit

I treated ingestion as a repeatable TD task. Python tools inspected the Alembic hierarchy, shading-group manifest, UV availability, material placements, and faceset coverage before changing the scene. The validated source contained 72 facesets and 112 mesh placements. Geometry and UVs were preserved while the assignments were mapped into clear material families.

MATERIAL FAMILY	LOOKDEV INTENT	PRIMARY CONTROLS
White paint	Aged coating that stays white	Seam dirt, lower dust, localized stains, roughness
Blue paint	Saturated painted metal	Coating roughness, sparse wear, protected color
Silver dome	Used aluminium	Reflection structure, micro-roughness, restrained grime
Mechanical metals	Readable dark and silver parts	Short-range grime, roughness separation
Lenses and indicators	Glass depth and practical emission	Material-specific isolation and light groups

Versioning strategy

Each major pass created a new Katana scene, a backup, and a frozen shader directory. Build, render, and validation were separate operations, which made failures traceable and prevented exploratory work from damaging an approved state. This was especially useful when the brief changed from a clean lookdev turntable to a full canyon shot.

Step 2 Camera and composition

I chose a 60 mm camera and a 2.35:1 frame to keep the character readable without turning the shot into a product render. The low amount of foreground clutter creates scale through negative space and lets the cast shadow lead the eye back to R2-D2.

The camera remained locked during material and lighting comparisons. Close reviews were produced as native high-resolution crop renders rather than moving the camera or enlarging a low-resolution image.

Step 3 Environment integration

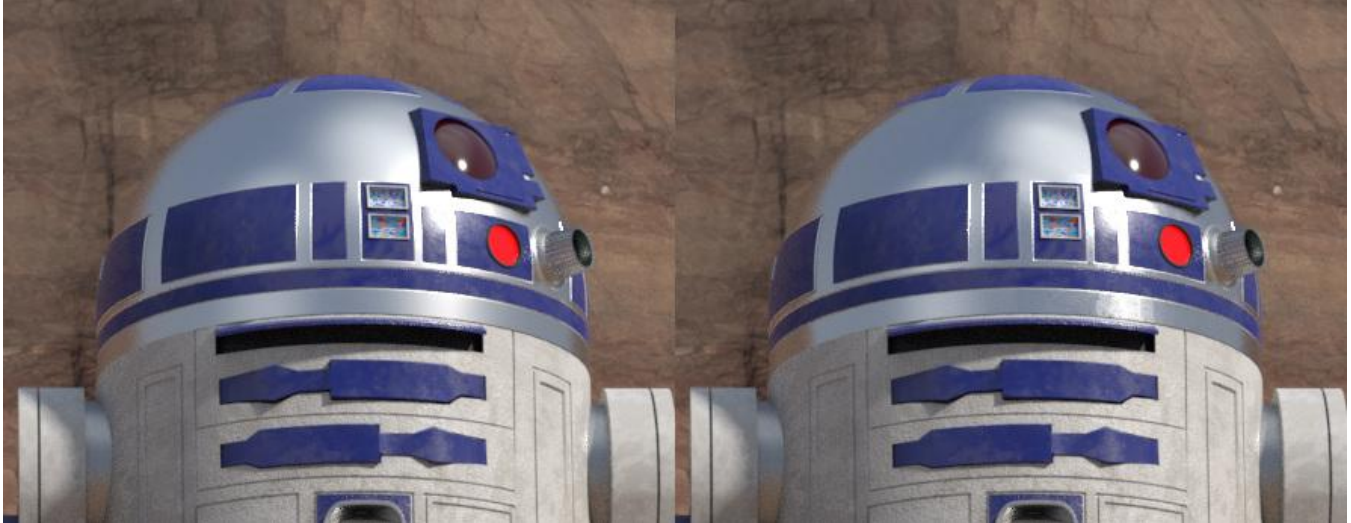
The environment combines cliff-side, gravelly-sand, and sandy-gravel texture families. Custom OSL nodes provide triplanar color, roughness, normal, and displacement response so the ground and canyon share a coherent world-space system without depending on perfect UVs.

An important art-direction choice was to remove the loose rock distribution. The individual stones looked procedural and competed with the silhouette. I retained geological detail in the canyon and moved realism into layered color, normal, roughness, and restrained displacement. The immediate area beneath the feet is protected from terrain displacement so contact remains stable.

Step 4 Material response

I evaluated every supplied 4K texture by map type, range, color space, and visual usefulness. Scalar sources were used where they improved response: aluminium imperfections for satin breakup, blue and painted-metal roughness for coating variation, a grunge glossiness map converted into irregular roughness, and mechanical roughness for localized grime.

Maps that introduced a rectangular decal edge, contained nearly flat displacement, duplicated an existing response, or had an unconfirmed normal convention were rejected. This kept the shader coherent and prevented arbitrary detail from replacing material logic.

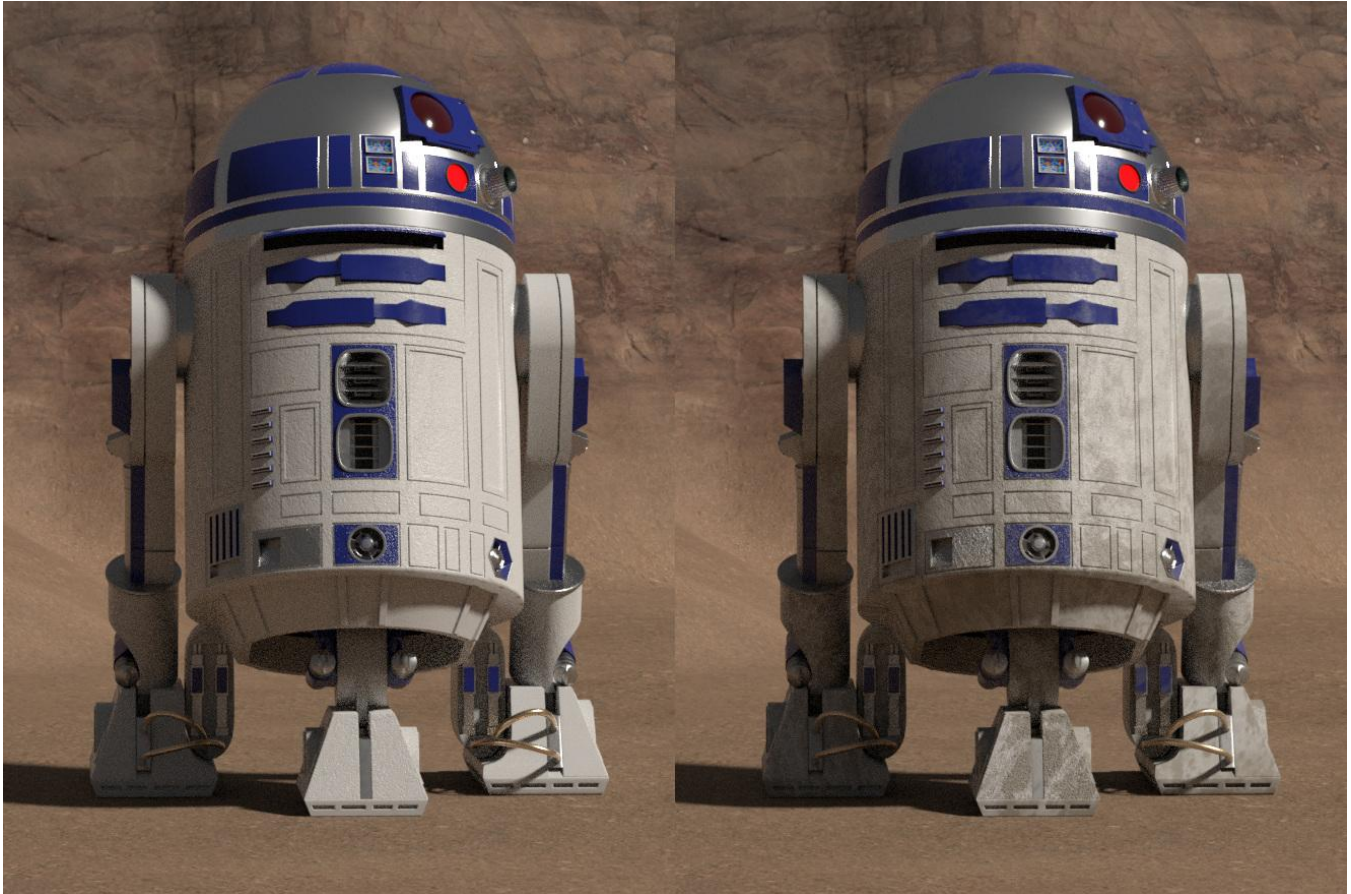


Dome comparison under matched lighting. The revised material adds reflection structure and fine breakup while preserving the blue panels.

Step 5 Procedural weathering

I authored a custom OSL weathering system with controls for panel dust, seam dirt, lower-body accumulation, mechanical grime, localized staining, dome wear, and micro-roughness. The masks combine world position, height, recess information, large and fine breakup, and art-directed stain anchors.

The key was hierarchy. Upper panels retain clean areas; the middle body receives irregular stains and stronger seam definition; the lower body, ankles, and feet carry the largest deposits. Dirt affects base color, roughness, and a very small amount of surface response instead of behaving like a flat overlay.



Matched physical renders. Left: cleaner material state. Right: localized desert weathering with white areas and blue identity preserved.

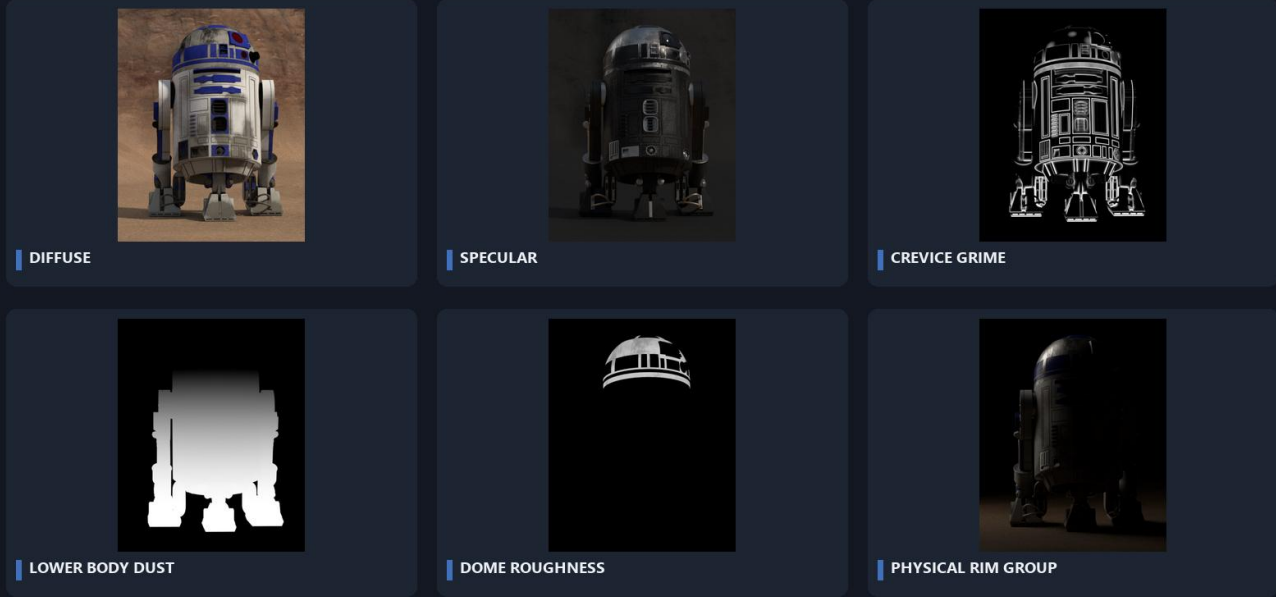
REGION	MEASURED ALBEDO RATIO	READING
Upper body	0.848	Largest amount of retained clean paint
Middle body	0.762	More staining and seam accumulation
Lower body	0.706	Stronger desert dust
Feet and ankles	0.681	Heaviest contact-zone exposure

The ratios compare the calibrated body-dirt pass with its previous state using the white-material albedo, independent of lighting. In the focused final pass, white-body dirt controls increased by 20 percent, dome dirt by 10 percent, and blue behavior remained unchanged.

Weathering masks and diagnostics

RENDERMAN DIAGNOSTICS

Lighting components and authored weathering masks from the final physical render



Selected RenderMan components and authored masks. The diagnostic package separates illumination, material response, wear, and subject separation.

Dedicated outputs made the look inspectable at render time. Crevice grime could be checked separately from large panel stains; lower-body dust could be evaluated independently of the light; and the dome micro-roughness mask could be compared directly with specular breakup. This reduced guesswork and kept Nuke from becoming the source of the weathering.

Controls exposed to lighting and lookdev

CONTROL	PURPOSE
panelDustAmount	Broad but irregular accumulation on painted panels
seamDirtAmount	Concentrated grime at borders, vents and recesses
lowerDustAmount	World-height accumulation toward the lower body and feet
localizedStainAmount	Art-directed breakup at selected body locations
domeDirtAmount	Lower-ring grime and controlled used-metal variation
mechanicalDirtAmount	Localized response on darker moving parts

Step 6 Lighting design

The lighting is motivated by a broad desert sun and environmental sky. A 4.6-degree distant source creates readable but softened cast-shadow edges. The sky carries cool and neutral information into shadow-side paint, while the warm ground and canyon contribute reflected color.

I strengthened the existing physically rendered rim contribution by 35 percent for the last pass. It appears mainly on the dome, shoulders, body sides, and upper legs, and remains in a dedicated light-group AOV for evaluation. The shape is broken by the real geometry; there is no continuous mask-based outline.

The small red circular indicator uses an emissive material with an HDR red core and a low-level red support light for nearby interaction. It remains a practical source rather than a red spotlight.



Native high-resolution crop render used to judge dome reflections, panel roughness, localized dirt, red emission, and rim placement.

LIGHT	FUNCTION	FINAL DECISION
Desert sun	Primary form and cast shadow	Warm, 4.6° angular extent, direction preserved
Environment sky	Shadow-side readability and reflections	Neutral fill with desert environment map
Canyon rim	Subject separation	Existing physical contribution increased 35%
Red practical support	Immediate local interaction	Low intensity; emission remains the visible source

Step 7 RenderMan output and denoise

The final physical render was produced at 1920 × 816 with pixel variance 0.012 and 12 to 128 samples. I rendered an additional 2880 × 1224 crop for upper-body review. RenderMan gen2_sym denoising used genuine variance, albedo, normal, normal variance, and sample-count guides; the raw beauty and variance data remain archived.

OUTPUT GROUP	CONTENTS	USE
Lighting	Diffuse and specular direct and indirect, emission, shadow	Balance and diagnose energy transport
Geometry data	Depth, normal, position	Measured focus, masks and diagnostics
Selections	Object and material Cryptomatte	Repeatable material- and asset-level isolation
Lookdev masks	Crevise grime, general dust, lower dust, paint breakup, dome roughness	Validate authored weathering independently
Light groups	Physical rim and red practical support	Evaluate separation and local interaction
Denoise guides	Variance, albedo, normal variance, sample count	Stable native denoising

Why the AOV design matters

The package let me answer whether a problem came from material response, illumination, geometry data, filtering, or comp. That separation became critical when an integration artifact appeared later in Nuke.



Native-pixel review of the feet and immediate ground, used to verify contact shadows, protected terrain displacement, and lower-body dust.

Step 8 Nuke integration and final finish

I kept the comp close to the RenderMan result. The workflow is divided into six versioned stages so each operation remains editable and can be re-rendered independently.

STAGE	OPERATION	KEY CONTROL
1	Subject and environment balance	Crypto-isolated values and restrained canyon reduction
2	Material response	White detail, blue saturation, aluminium highlights
3	Depth	Measured Z focus; maximum two-pixel distant defocus
4	Practical source	Emission-isolated 3 px and 8 px effect-only blooms
5	Film response	Warmth, HDR shoulder, diffusion, vignette and fine RGB grain
6	Output review	Linear EXR, no-grain variant, sRGB PNG and before-after

Cryptomatte drives independent controls for R2-D2, white paint, blue paint, the silver dome, mechanical parts, and the small red practical. The character and near-contact ground remain sharp. The emission source is isolated before blur and the two effect-only blooms are added once, avoiding duplicated illumination.



Same crop and display transform. Left: denoised RenderMan beauty. Right: final Nuke integration.

The final response uses a small warm bias in highlights and midtones, a logarithmic HDR shoulder, 2.2-pixel highlight diffusion at one-percent gain, a broad 5.5-percent vignette, and fine RGB grain. A no-grain version remains available for material review.

Step 9 Technical problem solving

Diagnosing a grading mask artifact

One Nuke test exposed a sawtooth pattern at the canyon-ground transition. I rendered diagnostic crops after each stage and traced the issue to a hard environment-region mask used for grading. The physical beauty and geometry were correct.

I copied the canyon region into a separate grading channel, feathered that control by 64 pixels, protected the robot matte, and added a cleaned depth ramp to the defocus mix. The original environment AOV stayed untouched. Re-rendering all downstream stages removed the artifact while retaining ground texture and foot contact.

Matching Viewer and export

I also found that image viewers handled a handful of non-opaque background pixels differently because of alpha association. The final EXR therefore retains the original alpha exactly, while the review PNG is intentionally opaque RGB. This makes the website image match the Nuke sRGB Viewer consistently.

Step 10 Automation and validation

Python automation handled scene inspection, Katana node and parameter updates, OSL compilation, RenderMan invocation, AOV packaging, staged Nuke graph creation, comparison generation, and validation reports. Automation supported the artistic process by shortening repetitive setup and making each iteration measurable.

CHECK	RESULT
Source integrity	287 protected files retained recorded hashes
Nuke baseline	Raw input matched the denoised RenderMan beauty exactly
Data fidelity	Alpha, AOV pixels, Cryptomatte IDs and manifests retained in final EXR
Depth safety	Zero filter delta on actor and near-contact region
Practical light	Localized bloom; saturated core without white clipping
Color	Scene-linear HDR EXR; sRGB transform applied once to review PNG

Result and relevance

The result is a desert shot in which R2-D2 reads as used, reflective, and grounded. The improvement comes primarily from physical material response, localized weathering, motivated lighting, reliable AOVs, and deliberate iteration. The Nuke pass supports those decisions without redesigning the render.

What this project demonstrates

- Visual judgment across material, lighting, environment, and comp.
- Scene-graph discipline and safe iteration on an existing asset.
- Shader and AOV design that makes physical decisions measurable.
- Render and color-pipeline troubleshooting across Katana, RenderMan, and Nuke.
- Automation that preserves artist control while making the workflow reproducible.

Credits and disclosure

This is a personal, noncommercial portfolio study. The R2-D2 model was an existing asset; my contribution begins with scene ingestion and includes the environment integration, look development, procedural weathering, lighting, rendering, compositing, and pipeline automation described above. R2-D2 and Star Wars are properties of Lucasfilm Ltd. Third-party texture sources should be credited on the published page according to their original licenses.